20-CARD DIGITAL NAMING PORTFOLIO

Part A: Structure → Name (A01–A10)
Part B: Name → Structure (B01–B10)

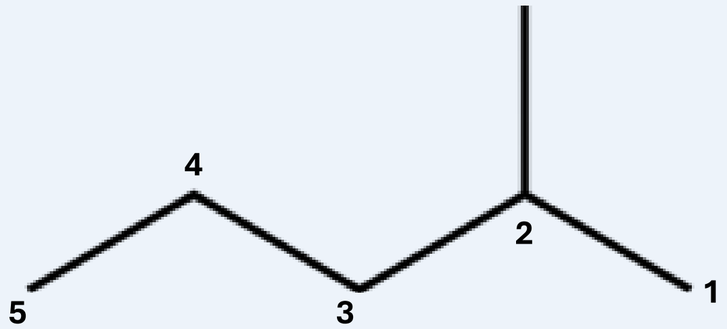
STUDENT DETAILS

Name: _Vhea Franz B. Aloba_____
Course/Section: _BSChE 3B_____
Instructor: __Engr. John Raymond Barajas_____

A01 — STRUCTURE → NAME

LEVEL: FOUNDATION • FORMULA COUNTED: C_6H_{14}

GIVEN STRUCTURE



NAMING TRAIL

Parent: 5 carbon atoms → pentane.

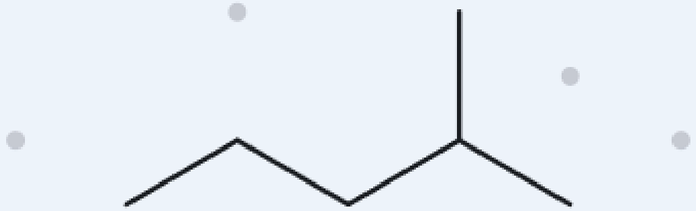
Numbering: methyl locant 2 vs 4; choose 2.

Substituent: one methyl group at C2.

MY IUPAC NAME

2-methylpentane

THE ALKANE CHECK



2-methylpentane

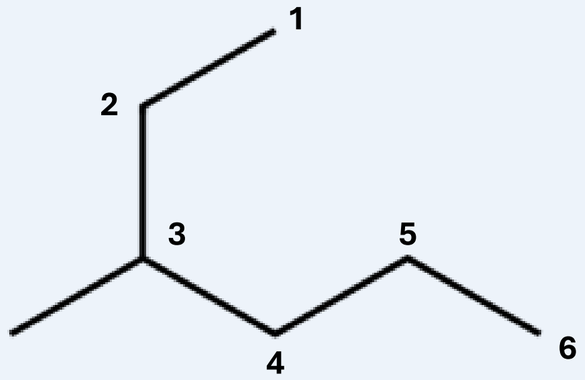
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A02 — STRUCTURE → NAME

LEVEL: FOUNDATION • FORMULA COUNTED: C_7H_{16}

GIVEN STRUCTURE



NAMING TRAIL

Parent: 6 carbon atoms → hexane.

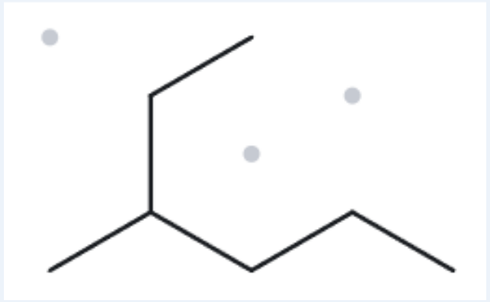
Numbering: one direction places the methyl group at C3; the opposite direction places it at C4. Choose C3.

Substituents: one methyl group at C3.

MY IUPAC NAME

3-methylhexane

THE ALKANE CHECK



3-methylhexane

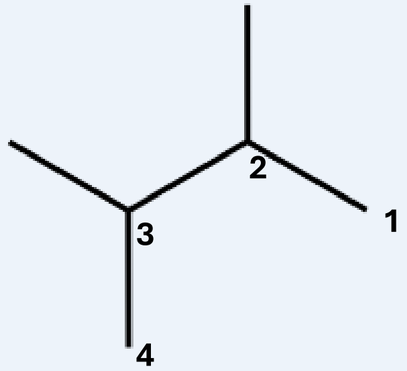
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A03 — STRUCTURE → NAME

LEVEL: FOUNDATION • FORMULA COUNTED: C_6H_{14}

GIVEN STRUCTURE



NAMING TRAIL

Parent: 4 carbon atoms → butane.

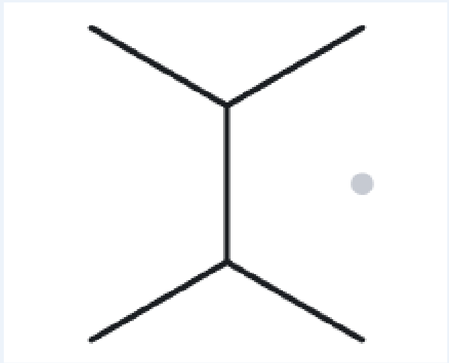
Numbering: the two methyl groups receive locants 2 and 3 from either direction.

Substituents: two methyl groups → dimethyl.

MY IUPAC NAME (WRITE BEFORE CHECKING)

2,3-dimethylbutane

THE ALKANE CHECK



2,3-dimethylbutane

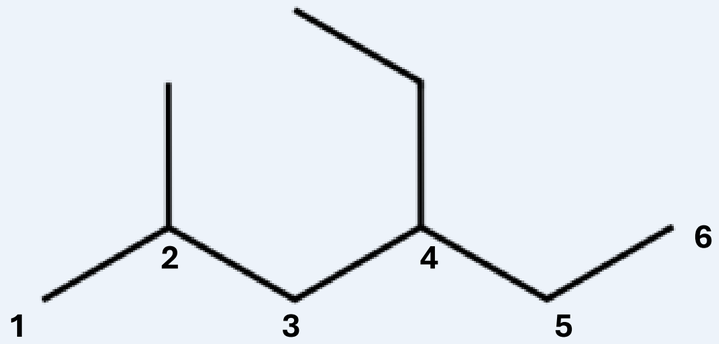
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A04 — STRUCTURE → NAME

LEVEL: DEVELOPING • FORMULA COUNTED: C_9H_{20}

GIVEN STRUCTURE



NAMING TRAIL

Parent: 6 carbon atoms → hexane..

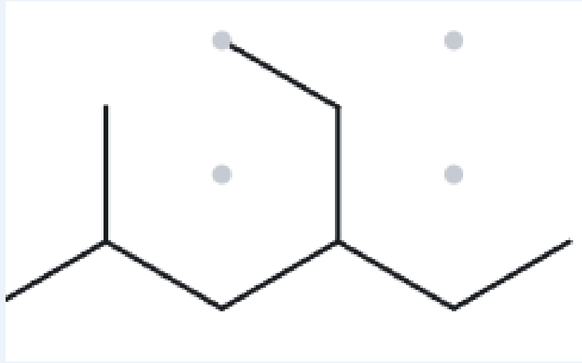
Numbering: compare both directions and use the lowest locant set; if the locant sets tie, apply the alphabetical tie-break.

Substituents:ethyl and methyl, listed alphabetically.

MY IUPAC NAME

4-ethyl-2-methylhexane

THE ALKANE CHECK



4-ethyl-2-methylhexane

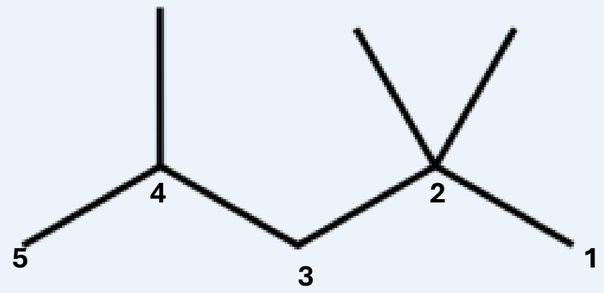
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A05 — STRUCTURE → NAME

LEVEL: DEVELOPING • FORMULA COUNTED: C_8H_{18}

GIVEN STRUCTURE



NAMING TRAIL

Parent: 5 carbon atoms → pentane..

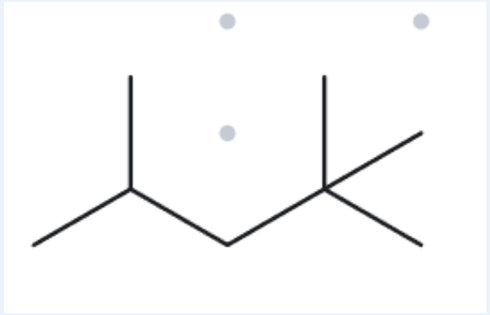
Numbering: tcompare the locant sets from both ends; choose the lowest set at the first point of difference.

Substituents: three methyl groups at C2, C2, and C4 → trimethyl..

MY IUPAC NAME

2,2,4-trimethylpentane

THE ALKANE CHECK



2,2,4-trimethylpentane

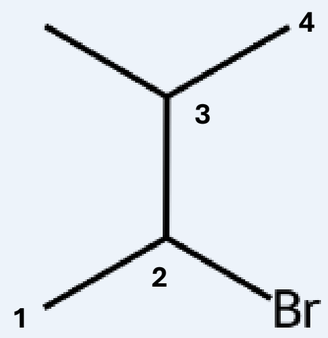
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A06 — STRUCTURE → NAME

LEVEL: DEVELOPING • FORMULA COUNTED: $C_5H_{11}Br$

GIVEN STRUCTURE



NAMING TRAIL

Parent: 4 carbon atoms → butane.

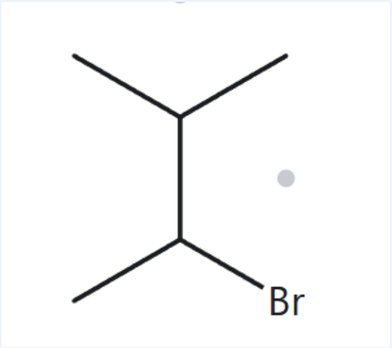
Numbering: both directions give the locant set 2,3. The alphabetical tie-break gives the lower number to bromo, which comes before methyl.

Substituents: bromo and methyl.

MY IUPAC NAME (WRITE BEFORE CHECKING)

2-bromo-3-methylbutane

THE ALKANE CHECK



2-bromo-3-methylbutane

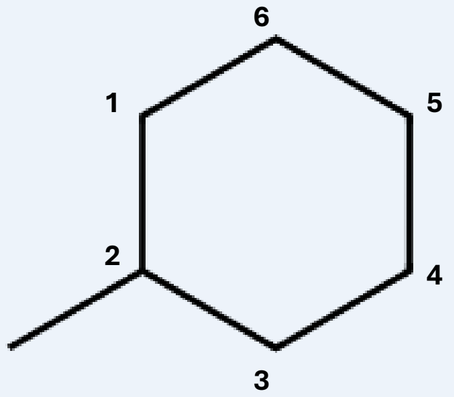
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A07 — STRUCTURE → NAME

LEVEL: DEVELOPING • FORMULA COUNTED: C_7H_{14}

GIVEN STRUCTURE



NAMING TRAIL

Parent: six-membered carbon ring → cyclohexane.

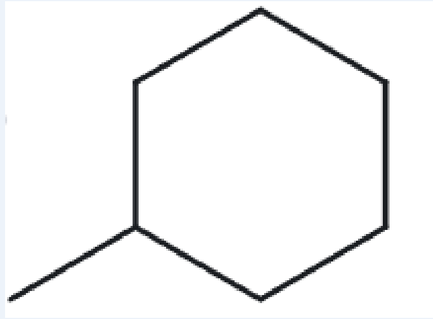
Numbering: a single substituent on the ring does not require a locant.

Substituents:one methyl group.

MY IUPAC NAME (WRITE BEFORE CHECKING)

methylcyclohexane

THE ALKANE CHECK



methylcyclohexane

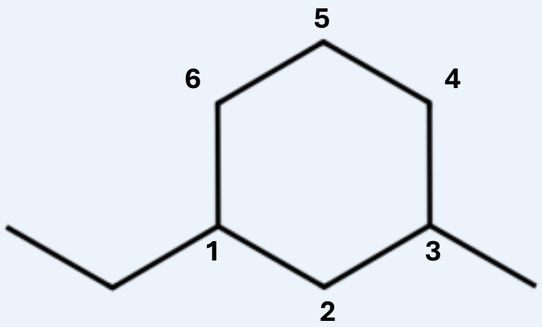
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A08 — STRUCTURE → NAME

LEVEL: CHALLENGE • FORMULA COUNTED: C_9H_{18}

GIVEN STRUCTURE



NAMING TRAIL

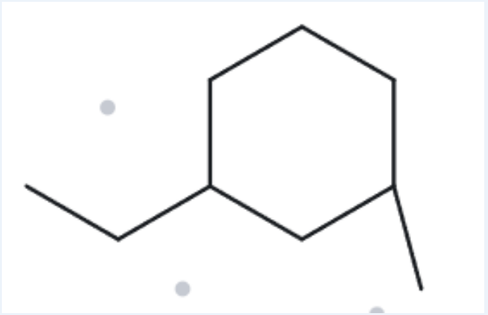
Parent: cyclohexane.
Numbering: the two substituents have locants 1 and 3 in either direction. Ethyl receives C1 because it comes alphabetically before methyl.

Substituents:ethyl and methyl.

MY IUPAC NAME (WRITE BEFORE CHECKING)

1-ethyl-3-methylcyclohexane

THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

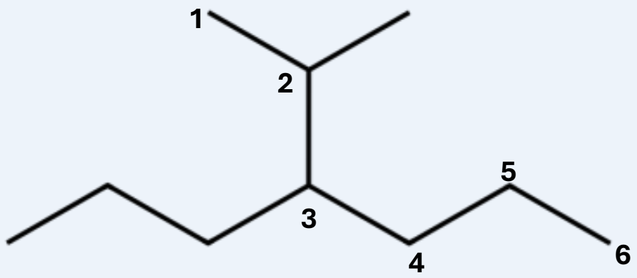
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A09 — STRUCTURE → NAME

LEVEL: CHALLENGE • FORMULA COUNTED: $C_{10}H_{22}$

GIVEN STRUCTURE



NAMING TRAIL

Parent: 6 carbon atoms → hexane.

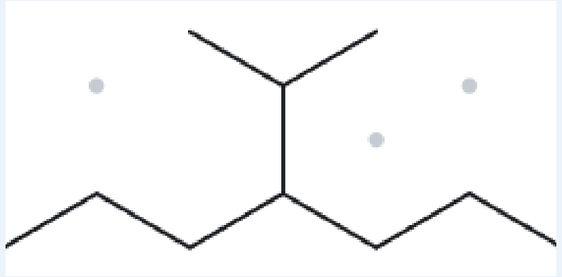
Numbering: Number the chain from the end nearest the substituents to obtain the lowest possible locants..

Substituents:One methyl group at C2 and one propyl group at C3.

MY IUPAC NAME (WRITE BEFORE CHECKING)

2-methyl-3-propylhexane

THE ALKANE CHECK



2-methyl-3-propylhexane

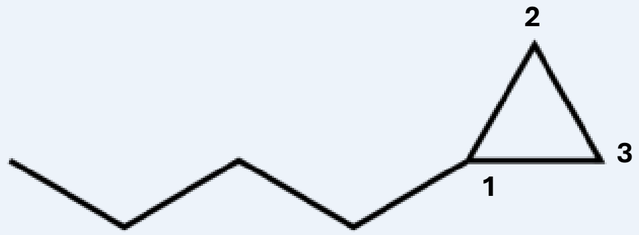
MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

A10 — STRUCTURE → NAME

LEVEL: CHALLENGE • FORMULA COUNTED: C7H14

GIVEN STRUCTURE



NAMING TRAIL

Parent: Cyclopropane ring (3 carbon atoms).

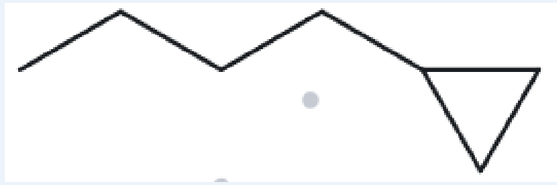
Numbering: The ring carbon attached to the butyl group is carbon 1. Since there is only one substituent on the ring, no additional locant is required in the name.

Substituents:One butyl group (4 carbon atoms) attached to carbon 1 of the cyclopropane ring.

MY IUPAC NAME (WRITE BEFORE CHECKING)

butylcyclopropane

THE ALKANE CHECK



butylcyclopropane

MATCH ✓

Verification: ✓ Longest chain confirmed · ✓ Lowest locant · ✓ Alphabetical order · ✓ Punctuation · ✓ Formula counted

B01 — NAME → STRUCTURE

LEVEL: FOUNDATION • GIVEN NAME: 2-methylpropane

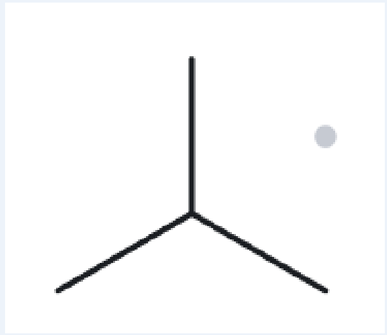
NAME DECODED

Parent: Propane → a 3-carbon chain.

Substituents: Methyl on C2.

Expected formula: 3 + 1 = 4 C; open-chain alkane → C_4H_{10}.

BOND-LINE STRUCTURE



2-methylpropane

FORMULA CHECK

Counted from the drawing: C_4H_{10}

Open-chain alkane: C_nH_{2n+2} = C_4H_{10}.

BACK-CHECK

Returned by The Alkane: 2-methylpropane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B02 — NAME → STRUCTURE

LEVEL: FOUNDATION • GIVEN NAME: 3-ethylpentane

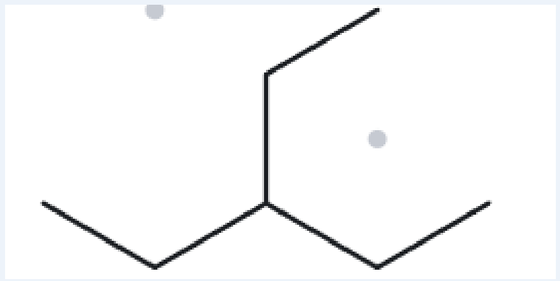
NAME DECODED

Parent: Pentane → a 5-carbon chain.

Substituents: Ethyl on C3

Expected formula: 5 + 2 = 7 C; open-chain alkane → C_7H_{16}.

BOND-LINE STRUCTURE



3-ethylpentane

FORMULA CHECK

Counted from the drawing: C_7H_{16}

Open-chain alkane: C_nH_{2n+2} = C_7H_{16}.

BACK-CHECK

Returned by The Alkane: 3-ethylpentane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B03 — NAME → STRUCTURE

LEVEL: FOUNDATION • GIVEN NAME: 2,2-dimethylbutane

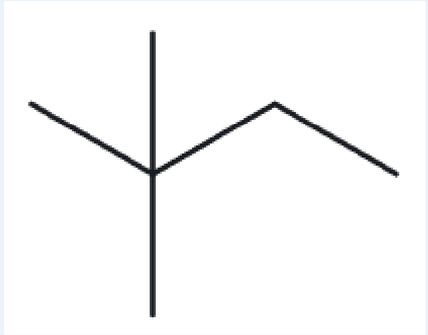
NAME DECODED

Parent: Butane → a 4-carbon chain.

Substituents: Two methyl groups on C2.

Expected formula: 4 + 1 + 1 = 6 C; open-chain alkane → C_6H_{14}.

BOND-LINE STRUCTURE



2,2-dimethylbutane

FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open-chain alkane: $C_nH_{2n+2} = C_6H_{14}$.

BACK-CHECK

Returned by The Alkane: 2,2-dimethylbutane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B04 — NAME → STRUCTURE

LEVEL: DEVELOPING • GIVEN NAME: 3-ethyl-2-methylpentane

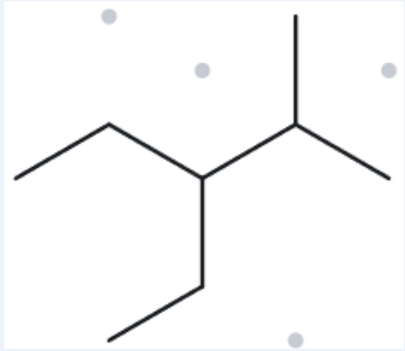
NAME DECODED

Parent: Pentane → a 5-carbon chain.

Substituents: Ethyl on C3; methyl on C2.

Expected formula: 5 + 2 + 1 = 8 C; open-chain alkane → C_8H_{18}.

BOND-LINE STRUCTURE



3-ethyl-2-methylpentane

FORMULA CHECK

Counted from the drawing: C_8H_{18}

Open-chain alkane: C_nH_{2n+2} = C_8H_{18}.

BACK-CHECK

Returned by The Alkane: 3-ethyl-2-methylpentane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B05 — NAME → STRUCTURE

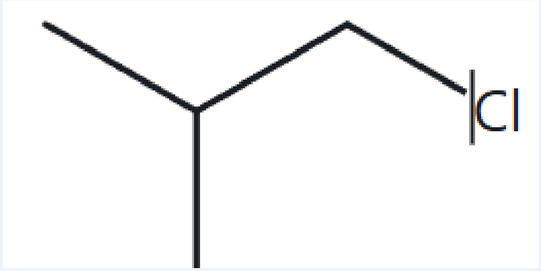
LEVEL: DEVELOPING • GIVEN NAME: 1-chloro-2-methylpropane

NAME DECODED

Parent: Propane → a 3-carbon chain.
Substituents: Chloro on C1; methyl on C2.

Expected formula: 3 + 1 = 4 C; open-chain alkane → C_4H_{10}; one H is replaced by Cl → C_4H_9Cl.

BOND-LINE STRUCTURE



1-chloro-2-methylpropane

FORMULA CHECK

Counted from the drawing: C_4H_9Cl

Open-chain alkane: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl.

BACK-CHECK

Returned by The Alkane: 1-chloro-2-methylpropane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B06 — NAME → STRUCTURE

LEVEL: DEVELOPING • GIVEN NAME: 1,2-dimethylcyclopentane

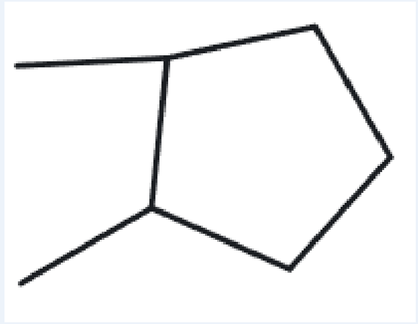
NAME DECODED

Parent: Cyclopentane → a 5-carbon ring.

Substituents: Methyl on C1 and methyl on C2.

Expected formula: 5 + 1 + 1 = 7 C; one ring → C_7H_{14}.

BOND-LINE STRUCTURE



1,2-dimethylcyclopentane

FORMULA CHECK

Counted from the drawing: C_7H_{14}

One ring: C_nH_{2n} = C_7H_{14}.

BACK-CHECK

Returned by The Alkane: 1,2-dimethylcyclopentane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B07 — NAME → STRUCTURE

LEVEL: DEVELOPING • GIVEN NAME: 4-ethyl-3,3-dimethylheptane

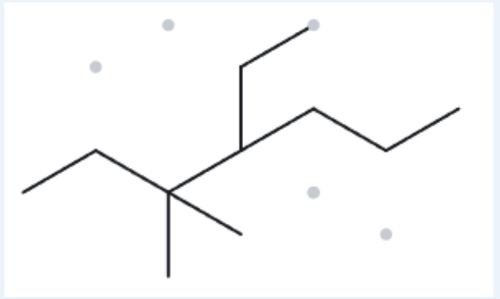
NAME DECODED

Parent: Heptane → a 7-carbon chain.

Substituents: Ethyl on C4; two methyl groups on C3.

Expected formula: 7 + 2 + 1 + 1 = 11 C; open-chain alkane → $C_{11}H_{24}$.

BOND-LINE STRUCTURE



4-ethyl-3,3-dimethylheptane

FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$

Open-chain alkane: C_nH_{2n+2} = $C_{11}H_{24}$

BACK-CHECK

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B08 — NAME → STRUCTURE

LEVEL: CHALLENGE • GIVEN NAME: tert-butylcyclohexane

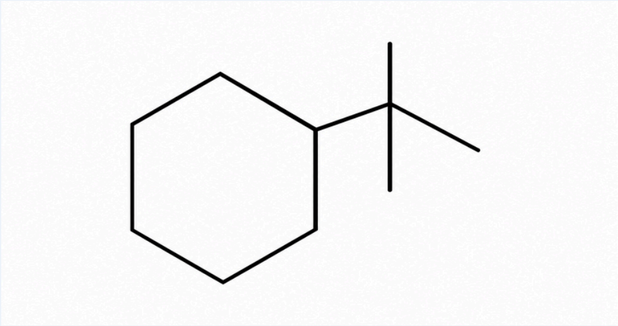
NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.

Substituents: One tert-butyl group attached to the ring. The tert-butyl group contains 4 carbon atoms.

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

BOND-LINE STRUCTURE



Tert-Butylcyclohexane

FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

Open-chain alkane: C_nH_{2n} = $C_{10}H_{20}$.

BACK-CHECK

Returned by The Alkane: tert-butylcyclohexane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B09 — NAME → STRUCTURE

LEVEL: CHALLENGE • GIVEN NAME: 3-ethyl-1,1-dimethylcyclohexane

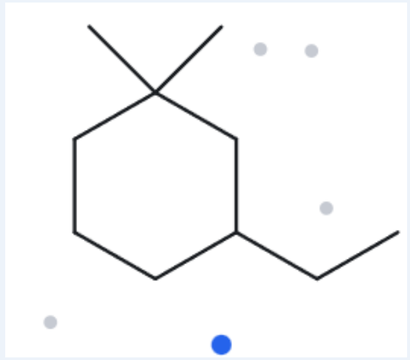
NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.

Substituents: Two methyl groups on C1; ethyl on C3.

Expected formula: 6 + 1 + 1 + 2 = 10 C; one ring → $C_{10}H_{20}$.

BOND-LINE STRUCTURE



3-ethyl-1,1-dimethylcyclohexane

FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

Open-chain alkane: C_nH_{2n} = $C_{10}H_{20}$.

BACK-CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B10 — NAME → STRUCTURE

LEVEL: CHALLENGE • GIVEN NAME: 2-ethylbutane (deliberately incorrect)

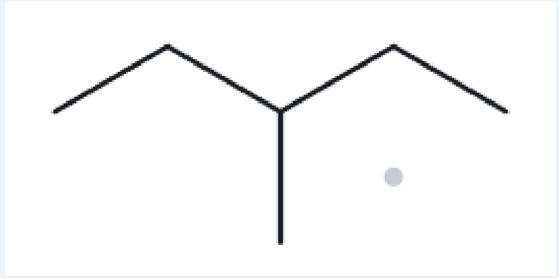
NAME DECODED

Parent: Butane → a 4-carbon chain.

Substituents: Ethyl on C2.

Expected formula: 4 + 2 = 6 C; open-chain alkane → C_6H_{14}.

BOND-LINE STRUCTURE



3-methylpentane

FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open-chain alkane: C_nH_{2n+2} = C_6H_{14}.

BACK-CHECK

Returned by The Alkane: 3-methylpentane

✓ **Check:** Same as given name.

Verification: ☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

PRE-SUBMISSION CHECKLIST

Complete these checks before exporting the final portfolio PDF.

□ 20 cards completed in order: A01–A10, then B01–B10

□ Part A names written before checking in The Alkane

□ Part A structures copied from the provided image pack and parent carbons numbered

□ All Part B structures built and checked in The Alkane

□ B10 drawn exactly as written, then corrected name and broken rule recorded

□ 20 cropped The Alkane screenshots saved using card-code filenames

□ Names use correct commas and hyphens with no spaces

□ AI disclosure lists The Alkane and any AI tool used

□ Editable PPTX retained and one complete PDF exported

□ Public GitHub repository verified; link submitted through Google Classroom